



CL-2001 Data Structures Lab # 7

Objectives:

- Recursion
- Array based Trees

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function about its functionality.
3. Use understandable name of variables.
4. Proper indentation of code is essential.
5. Write a C++ statement(s) for each of the following task one after the other, in the same order.
6. Make a Microsoft Word file and paste all of your C++ code with all possible screenshots of **every task output in MS word and submit .cpp file with word file.**
7. Make separate .cpp files for all tasks and use this format **22F-1234_Task1.cpp.**
8. First think about statement problems and then write/draw your logic on copy.
9. After copy pencil work, code the problem statement on MS Studio C++ compiler.
10. At the end when you done your tasks, attached C++ created files in MS word file and make your submission on Google classroom. (Make sure your submission is completed).
11. Please submit your word file in this format **22F-1234_L1.docx**
12. Do not submit your assignment **after the deadline.**
- 13. Do not copy code from any source otherwise you will be penalized with negative marks.**

Problem: 1 | Recursion

Write a recursive function that receives an integer consisting of any number of digits. Your function should calculate and return the summation and average of the integer digits.

Enter number: 5

Total sum: 15

Average : 3

Note: Use minimum lines of code as possible to get max marks.

Problem: 2 | Recursion

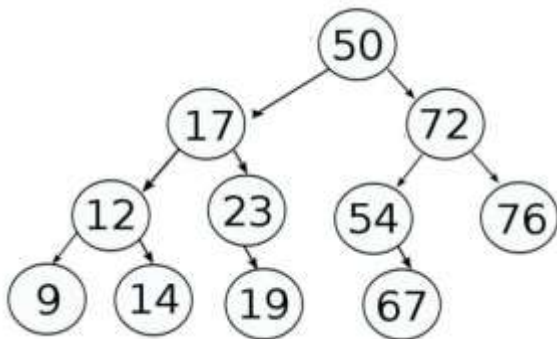
Implement Binary Search using recursion using that function:

- `int binarySearch(int arr[], int target, int low, int high)`

Note: Use minimum lines of code as possible to get max marks.

Problem: 3 | Queue Linked List

Consider the given binary tree and answer the following questions.



1. Total leaf nodes?
2. Siblings of 12?
3. Ancestors of 17?
4. Descendent of 23?
5. Height and depth of given tree from 54?
6. Is it a complete binary tree? Total number of nodes in a complete binary tree with depth 4 will be?
7. Pre-order traversal will be?
8. In-order traversal will be ?



9. Post-order traversal will be?

Problem: 4 | Binary Tree Array based

Implement Binary tree ADT. Create following functions:

1. isEmpty()
2. isFull()
3. getRoot()
4. insert(data, parent)
5. delete(data)
6. pre_order_display()
7. Height_of_tree()



Best of luck